CHAPTER 1 · LESSON 2 OF 5

Types of Data

BY THE END

You can classify data as structured, semi-structured, or unstructured, and as quantitative or qualitative.

Overview

A shop owner opens a spreadsheet of last week's sales and sees numbers, product names, and customer comments all mixed together. Some entries fit neatly in columns; others are long notes in a single cell.

last-week-sales.csv

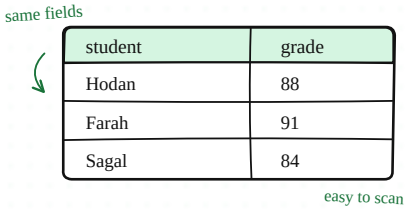
#	Product name	Number	Customer comment
1	Rice	24	“Great service”
2	Cooking oil	12	A longer note fills this cell…

Before choosing how to analyze this, you need to know what *type* of data you are looking at.

Structured vs unstructured data

Structured data

Structured data is organized in a clear, predictable format. It usually fits neatly into rows and columns, like a table.

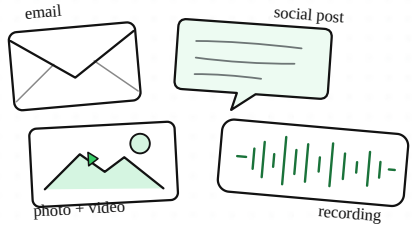


EXAMPLES

- A spreadsheet of student names and grades
- A database table of customer orders
- A CSV file of product prices

Unstructured data

Unstructured data does not follow a fixed format. It is harder to organize automatically.



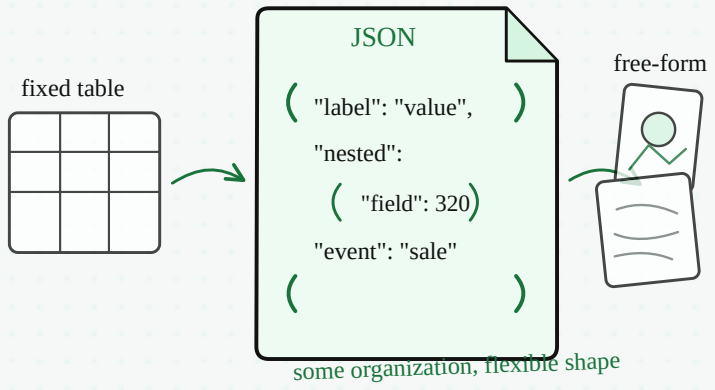
EXAMPLES

- Emails
- Photos and videos
- Social media posts
- Call center recordings

Semi-structured data

Semi-structured data sits between the two. It has some organization, but not a fixed table layout. **JSON** is a common example. It uses labels and nested fields, which makes it useful for apps and APIs but less rigid than a database table.

```
{ }  labels + nested fields
     flexible, not fixed
```



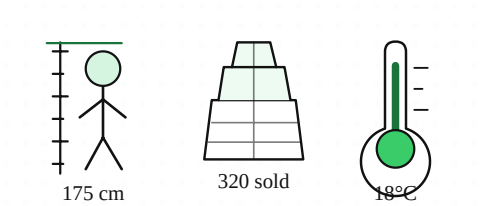
Most real-world projects use a mix of all three. A company might store order details in a structured database, customer reviews as unstructured text, and app event logs as semi-structured JSON.

- **order details** — structured database
- **customer reviews** — unstructured text
- **app event logs** — semi-structured JSON

Quantitative vs qualitative data

Quantitative data

Quantitative data can be counted or measured with numbers.

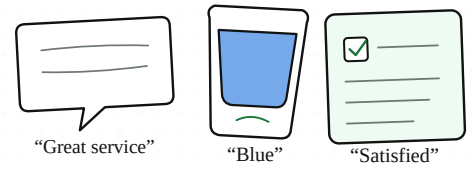


EXAMPLES

- Height: `175 cm`
- Number of products sold: `320`
- Temperature: `18°C`

Qualitative data

Qualitative data describes qualities or categories. It often comes in words or labels.



EXAMPLES

- Customer feedback: “Great service”
- Product color: “Blue”
- Survey response: “Satisfied”

Labels like product color or satisfaction level are often called **categorical data**. They group items into categories rather than measure them with numbers.

Why data types matter

Knowing the type of data helps you answer questions like:

Can I calculate an average?

Can I make a chart from this?

Do I need text analysis tools?

Is this data ready to use, or does it need cleaning?

For example, you can average **quantitative sales numbers**, but you cannot average qualitative comments like “good” and “excellent” in a meaningful way without extra processing.

SALES NUMBERS

`average` ✓

VS

COMMENTS

“good” + “excellent”